

AI Tool Usage Declaration

As per the integrity requirements of the competition, this document describes the use of AI tools while developing the UrbanTransit IQ project. No predictive AI services are invoked at runtime; rather all predictions, classifications, clusters, and recommendations that the system makes are generated using Spark MLlib, scikit-learn, and rule-based algorithms

Field	Content
Tool	Claude
Purpose	Coding assistant
Type of help	Scaffolding, boilerplates, debugging of Windows/Spark configuration, documentation writing, later code restructuring
Files / modules affected	src/utiq/ (simulate, pipeline, analysis, models, advice), dashboard/, docs/, tests/
Modifications made	Adjusted datasets size according to spec minimums; solved Windows Spark problem (HADOOP_HOME/winutils, PYSPARK_PYTHON, pandas 3 vs PySpark 4.2); optimized queries using anti-join instead of isin(); adjusted hardcoded values to settings.py; implemented the leak prevention measure; separated CSS/JS from the dashboard code
Testing done	Re-created the dataset and verified that it is indeed a perfect match; executed all 17 processes end-to-end; 285 tests, of which 45 require a Spark session, related to band limits, leakage guard, model registry versioning, runs and audits, data and model quality gates.
Verified by	Muhammad Hamza Mughal

Not AI-generated: the data set is taken from a seeded simulator, not from an LLM. Predictions for overcrowding and delays, forecasting of demand, route clustering, and all recommendations are generated through Spark MLlib, scikit-learn, and rule-based code within the project there is no call to any AI-generating API.